

Basic Electrical Engineering

MODULE 3 — PART 2: QUESTION BANK

Module 3 (6 hrs): Basics of Magnetic Circuits and DC Machines

For First-Year Students — with complete answers and diagrams

Course: Basic Electrical Engineering · Course Code: EE100

Question Paper Structure

Section	Marks per Question	Scope	Questions
Section A	2 marks	Compulsory — from every module	Q1 - Q36
Section B	7 marks	Compulsory — from every module	Q1 - Q14
Section C	Numericals	Solved problems (all topics)	N1 - N13

- Every section covers **all the topics** of Module 3 — no topic is left out.
- No question is repeated between sections.
- Section C gives every numerical problem solved step by step, with its diagram.

SECTION A — 2-Mark Questions

Q1. What is a magnetic field and a magnetic force?

Answer: The space around the poles of a magnet is the magnetic field. The force of attraction or repulsion between two magnets is the magnetic force; unlike poles attract and like poles repel.

Q2. Define magnetic flux and give its unit.

Answer: The number of magnetic lines of force set up in a magnetic circuit is the magnetic flux ϕ ; its unit is the weber (Wb). It always forms closed loops and never intersects itself.

Q3. What is magnetic flux density and what is its unit?

Answer: It is the flux passing through an area perpendicular to the direction of flux, $B = \phi/A$. Its unit is the tesla (T) or Wb/m².

Q4. What is an electromagnet? Give one advantage.

Answer: A magnet in which the magnetic field is produced by an electric current. Its advantage is that the field can be changed quickly by controlling the current.

Q5. What is the nature of the magnetic field around a straight current-carrying conductor?

Answer: It consists of concentric circles centred on the axis of the conductor, with density proportional to the current; the direction is given by the right-hand grip rule.

Q6. State Maxwell's right-hand grip rule.

Answer: Hold the conductor in the right hand so the fingers wrap around it and the thumb points along the current; the fingers then show the direction of the circular magnetic field lines.

Q7. What is a solenoid?

Answer: A current-carrying wire wound spirally in the form of a helix about an axis. It acts like a bar magnet with north and south poles.

Q8. Define magnetomotive force (MMF) and give its unit.

Answer: The MMF is the cause of magnetic flux in a magnetic circuit, $F = NI$. Its unit is the ampere-turn (AT).

Q9. Define magnetic field strength (magnetising force) and give its unit.

Answer: It is the magnetomotive force per unit length of the flux path, $H = NI/l$. Its unit is AT/m.

Q10. What is permeability? Write the relation for B.

Answer: Permeability is the property of a medium that decides the flux density for a given magnetising force; $\mu = \mu_0\mu_r$ and $B = \mu H$.

Q11. What is relative permeability?

Answer: The ratio of the permeability of a medium to the permeability of free space, $\mu_r = \mu/\mu_0$. It is 1 for air and much greater than 1 for iron.

Q12. Define reluctance and give its unit.

Answer: The opposition offered by a magnetic circuit to the magnetic flux, $S = l/(\mu_0\mu_r A)$. Its unit is AT/Wb.

Q13. State the analogy between magnetic and electric circuits.

Answer: MMF $\leftrightarrow$ emf, flux $\leftrightarrow$ current, reluctance $\leftrightarrow$ resistance and permeability $\leftrightarrow$ conductivity.

Q14. What is a composite series magnetic circuit?

Answer: A magnetic circuit made of two or more parts in series (e.g. iron and an air-gap). The same flux passes through all parts, so the total reluctance is $S = S_1 + S_2 + \dots$.

Q15. What is a B-H curve?

Answer: The graph of flux density B against magnetising force H. It has a linear region, a knee point and a saturation region.

Q16. What is a hysteresis loop? Define retentivity and coercivity.

Answer: The loop traced by B when H is taken through a complete cycle. Retentivity is the flux density left when H is zero; coercivity is the reverse H needed to bring B to zero.

Q17. State Faraday's law of electromagnetic induction.

Answer: The induced emf is directly proportional to the rate of change of flux linkages, $e = -N(d\phi/dt)$.

Q18. State Lenz's law.

Answer: The direction of the induced emf is such that the current it produces sets up a magnetic field opposing the change of flux that produced it.

Q19. What is a dynamically induced emf? Give its formula.

Answer: The emf induced when a conductor moves in a magnetic field: $e = B l v \sin \theta$, its direction given by Fleming's right-hand rule.

Q20. What is a statically induced emf? Name its two types.

Answer: The emf induced when the conductor and field are stationary but the linking flux changes; the two types are self-induced and mutually induced emf.

Q21. Classify DC machines.

Answer: A DC generator converts mechanical energy into electrical energy, and a DC motor converts electrical energy into mechanical energy.

Q22. State the principle of operation of a DC generator.

Answer: Whenever flux is cut by a conductor, an emf is induced; if the conductor circuit is closed, a current flows. The relative motion is obtained by rotating conductors in a constant DC field.

Q23. What is the function of the commutator?

Answer: It is a mechanical rectifier that converts the alternating emf in the armature into direct voltage in the external circuit by reversing the coil connections at the instant the current reverses.

Q24. What is the function of the yoke in a DC machine?

Answer: It is the outer frame that protects the machine, supports the poles and carries the magnetic flux.

Q25. What are the pole core and pole shoe?

Answer: The pole core holds the pole shoe on the yoke, and the larger pole shoe spreads the flux over the air-gap to reduce reluctance.

Q26. What is the function of the field winding?

Answer: Copper coils on the pole cores; the DC field current makes adjacent poles of opposite polarity and produces the field flux. It is a stationary part.

Q27. Why is the armature core laminated?

Answer: The armature core is built of laminations to reduce the eddy-current loss. It also gives a low-reluctance path to the flux.

Q28. What is the armature winding and what does it do?

Answer: The winding placed in the armature slots; it is the winding in which the emf is induced. Its conductors are connected in series to increase voltage and in parallel paths to increase current.

Q29. What are brushes and of what material are they made?

Answer: Brushes collect current from the commutator; they are usually made of carbon or graphite and housed in brush holders.

Q30. Compare lap and wave windings.

Answer: Lap winding has $A = P$ parallel paths and suits high-current, low-voltage machines; wave winding has $A = 2$ parallel paths and suits high-voltage, low-current machines.

Q31. Write the EMF equation of a DC generator and explain the symbols.

Answer: $E = \frac{\phi Z N P}{60 A}$ volts, where ϕ = flux per pole, Z = total conductors, N = speed in rpm, P = poles and A = parallel paths ($A = P$ for lap, $A = 2$ for wave).

Q32. Name the types of DC generator.

Answer: Separately excited, series, shunt, and compound (cumulative and differential) generators.

Q33. State the applications of a series generator.

Answer: Used as boosters on DC feeders and as a constant-current generator for welding and arc lamps.

Q34. What is the force on a current-carrying conductor? State Fleming's left-hand rule.

Answer: A force $F = B I l$ acts on the conductor; its direction is given by Fleming's left-hand rule — with the forefinger along the field and the middle finger along the current, the thumb gives the force.

Q35. State the principle of a DC motor and define torque.

Answer: Rotor conductors carrying current in the air-gap field each experience a force, producing torque at the circumference so the rotor rotates. Torque is the moment of a force causing rotation.

Q36. What is back emf in a DC motor?

Answer: The emf induced in the rotating armature that opposes the applied voltage; $E_b = V - I_a R_a$.

SECTION B — 7-Mark Questions**Q1. Explain the basic magnetic quantities — magnetic field, flux, flux density and the magnetic force — with examples.**

Answer:

The space around the poles of a magnet is the magnetic field, represented by lines of force. The magnetic force between two magnets follows Coulomb's first law: unlike poles attract, like poles repel. The number of lines of force is the magnetic flux ϕ (weber); it forms closed loops and never intersects itself. The flux per unit perpendicular area is the flux density $B = \phi/A$, measured in tesla. For example, a pole face of 200 mm × 100 mm with 150 μWb has $B = 7.5 \text{ mT}$.

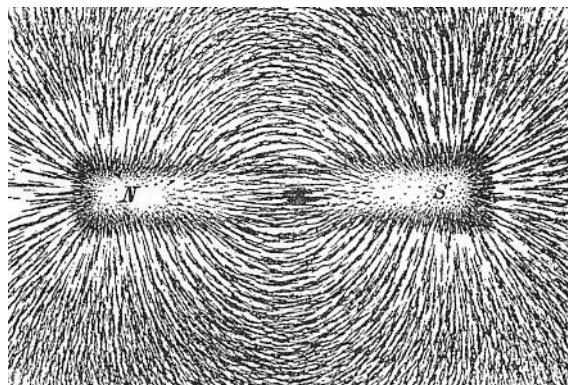


Figure 1: Magnetic field of a bar magnet.

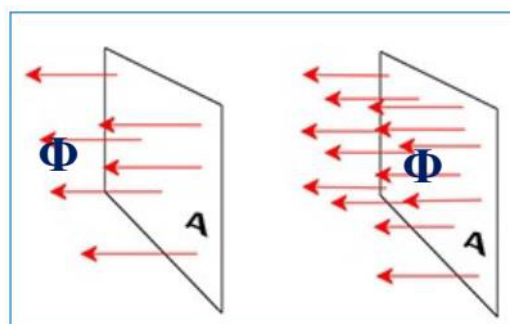


Figure 2: Flux lines form closed loops.

Q2. Explain electromagnets, the magnetic field of a current-carrying conductor, and the solenoid.

Answer:

An electromagnet produces its field by an electric current and its field can be changed by controlling the current. A straight current-carrying conductor has concentric circular field lines whose density is proportional to the current, the direction being given by Maxwell's right-hand grip rule. A wire wound into a helix is a solenoid, which behaves like a bar magnet.

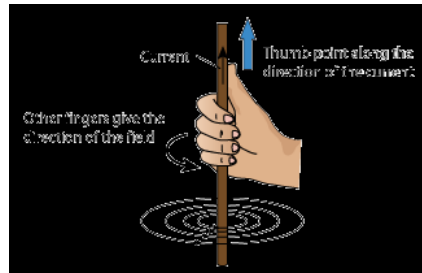


Figure 3: Maxwell's right-hand grip rule.

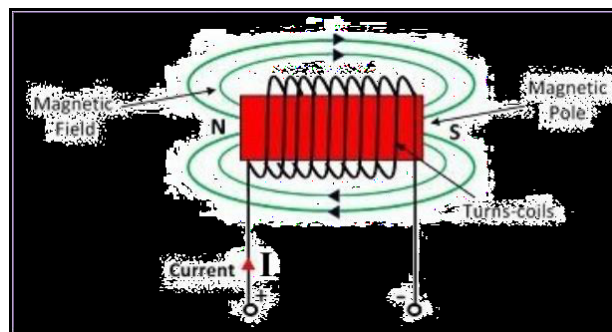


Figure 4: A solenoid acts like a bar magnet.

Q3. Define MMF, magnetic field strength and permeability, and derive the relation between them.

Answer:

The MMF is the cause of flux, $F = N I$, in ampere-turns. The magnetising force is MMF per unit length, $H = NI/l$, in AT/m. The permeability of free space is $\mu_0 = 4\pi \times 10^{-7}$ H/m and for any medium $\mu = \mu_0 \mu_r$. The flux density is related by $B = \mu H$.

$$F_m = N \times I \text{ (AT).}$$

Figure 5: MMF of a coil.

$$H = \frac{mmf}{l}$$

Figure 6: Magnetising force.

Q4. Define reluctance, give its formula, and explain the analogy with the electric circuit.

Answer:

Reluctance is the opposition offered to magnetic flux: $S = l/(\mu_0\mu_r A)$, in AT/Wb. The magnetic circuit is analogous to the electric circuit: MMF $\leftrightarrow$ emf, flux $\leftrightarrow$ current, reluctance $\leftrightarrow$ resistance, permeability $\leftrightarrow$ conductivity. Thus flux = MMF/reluctance, just as current = emf/resistance.

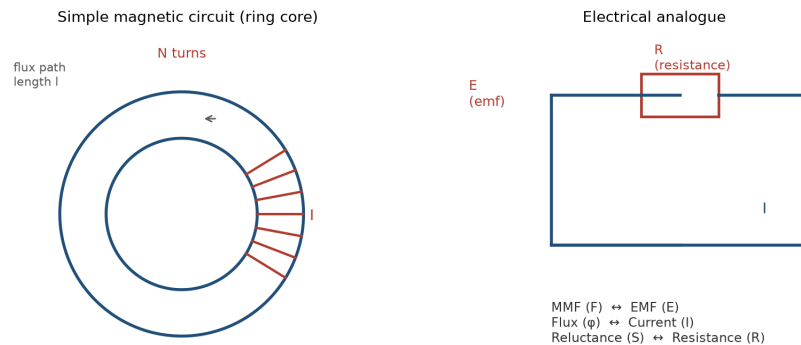


Figure 7: Magnetic circuit and its electrical analogue.

Q5. Explain composite series magnetic circuits and how the current for a required air-gap flux is found.

Answer:

In a composite series circuit (e.g. iron with an air-gap) the same flux passes through all parts, so the total reluctance is $S = S_1 + S_2 + \dots$. The required MMF is $F = \phi S$ and the coil current is $I = F/N$. The air-gap has a very high reluctance and usually dominates the calculation.

$$l = 2\pi r = 2 \times \pi \times 50 \times 10^{-3} \text{ m}; A = 400 \times 10^{-6} \text{ m}^2; I = 0.5 \text{ A};$$

$$\Phi = 0.1 \times 10^{-3} \text{ Wb}; \mu_r = 200$$

$$(a) \text{ Reluctance } S = \frac{l}{\mu_0\mu_r A} = \frac{2 \times \pi \times 50 \times 10^{-3}}{(4\pi \times 10^{-7})(200)(400 \times 10^{-6})}$$

$$= 3.125 \times 10^6 / \text{H}$$

Figure 8: A composite series magnetic circuit.

Q6. Explain the B-H curve and the hysteresis loop with their important points.

Answer:

The B-H curve shows how B varies with H: a linear region at low H, a knee point, then saturation where B hardly increases. Taking H through a complete cycle gives the hysteresis loop. Retentivity B_r is the flux density remaining at $H = 0$ and coercivity H_c the reverse H needed to bring B to zero. The loop area represents the hysteresis loss per cycle.

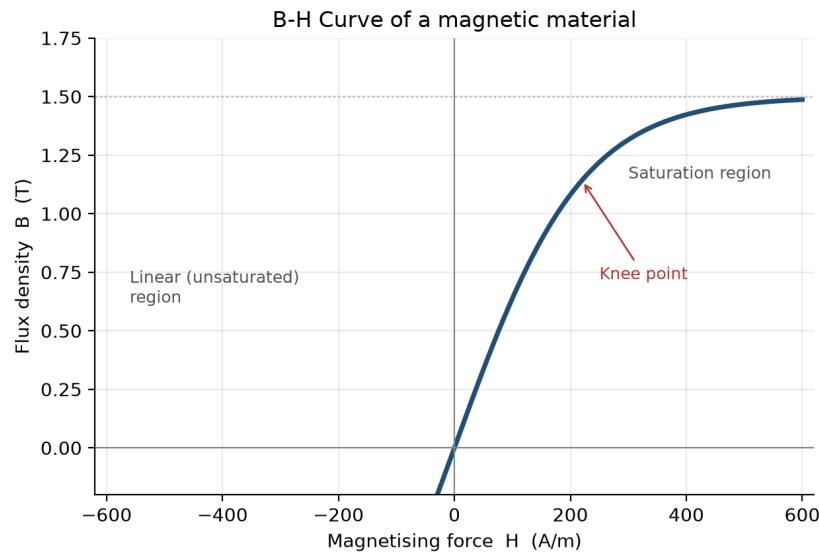


Figure 9: The B-H curve.

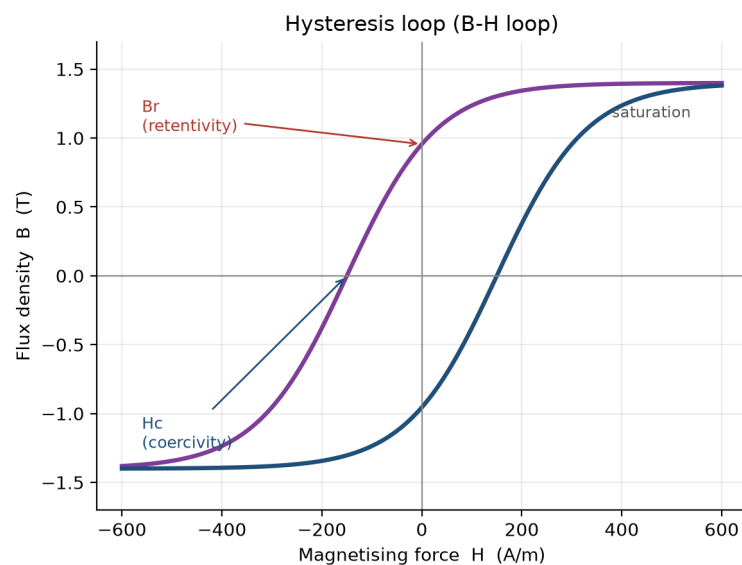


Figure 10: The hysteresis loop.

Q7. Explain electromagnetic induction: Faraday's law, Lenz's law, and the types of induced emf.

Answer:

Faraday's law states $e = -N(d\phi/dt)$: the induced emf is proportional to the rate of change of flux linkages. Lenz's law gives the direction — the induced emf opposes the change of flux. Emf may be dynamically induced (conductor moving in a field, $e = Blv \sin\theta$, Fleming's right-hand rule) or statically induced (self-induced and mutually induced, when the linking flux changes without motion).

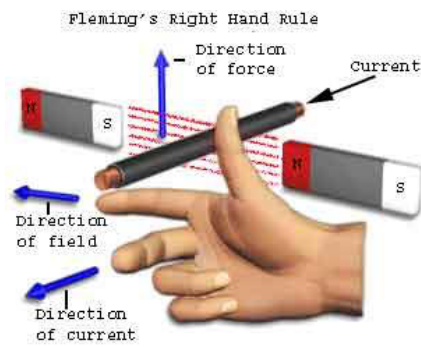


Figure 11: Fleming's hand rule.

Q8. Explain the construction of a DC machine with the function of each part.

Answer:

The main parts are the yoke (frame that supports the poles and carries flux), the pole cores and pole shoes (spread the flux), the field winding (produces the field flux), the laminated armature core (low-reluctance path, reduces eddy-current loss), the armature winding (where emf is induced), the commutator (mechanical rectifier) and the brushes (collect current).

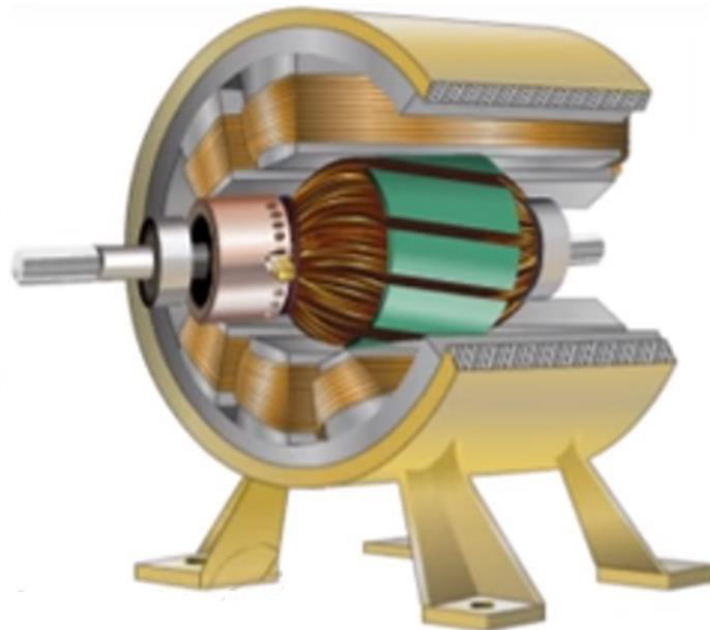


Figure 12: Construction of a DC machine.

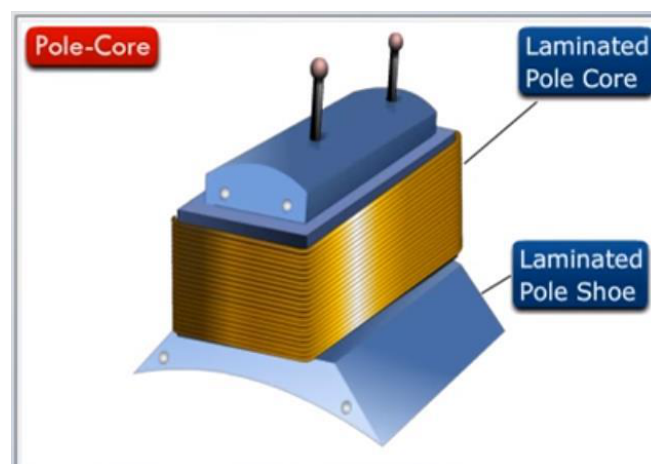


Figure 13: Pole core and pole shoe.

Q9. Explain the working principle of a DC generator and the action of the commutator.

Answer:

A DC generator works on the principle that when flux is cut by a conductor an emf is induced. The field is constant DC and the conductors rotate. The emf generated in the loop is alternating, and the commutator — a mechanical rectifier — reverses the coil connections to the load at the instant the coil current reverses, so the load current is unidirectional.

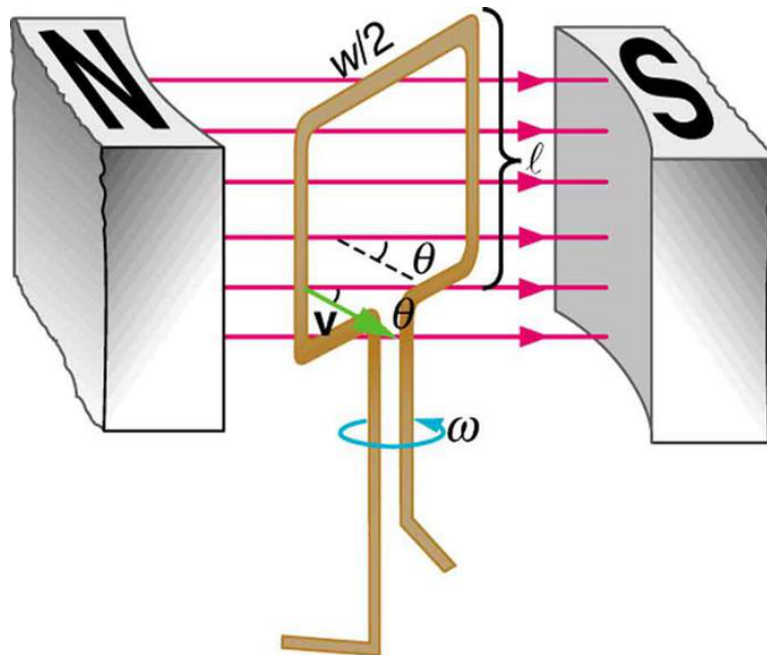


Figure 14: Working principle of a DC generator.

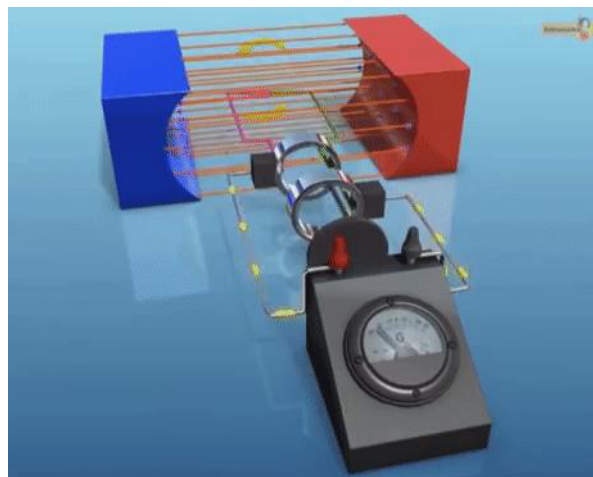


Figure 15: Emf in the rotating loop.

Q10. Derive the EMF equation of a DC generator and explain each term.

Answer:

The average emf per conductor is $\phi PN/60$ and the number of conductors in series per parallel path is Z/A , so the generated emf is:

$$E = (\phi Z N P) / (60 A) \text{ volts}$$

where ϕ = flux per pole (Wb), Z = total armature conductors, N = speed in rpm, P = number of poles and A = number of parallel paths ($A = P$ for lap, $A = 2$ for wave).

$$\begin{aligned}
 &= \text{No. of slots} \times \text{No. of conductors/slot} \\
 P &= \text{No. of generator poles} \\
 A &= \text{No. of parallel paths in armature} \\
 N &= \text{armature rotation in revolutions per minute (r.p.m.)} \\
 E &= \text{e.m.f. induced in any parallel path in armature} \\
 \text{Generated e.m.f. } E_g &= \text{e.m.f. generated in any one of the parallel paths i.e. } E. \\
 \text{Average e.m.f. generated/conductor} &= \frac{d\Phi}{dt} \text{ volt } (\because n = 1) \\
 \text{Now, flux cut/conductor in one revolution } d\Phi &= \Phi P Wb \\
 \text{No. of revolutions/second} &= N/60 \therefore \text{Time for one revolution, } dt = 60/N \text{ second} \\
 \text{Hence, according to Faraday's Laws of Electromagnetic Induction,} \\
 \text{E.M.F. generated/conductor} &= \frac{d\Phi}{dt} = \frac{\Phi P N}{60} \text{ volt}
 \end{aligned}$$

Figure 16: EMF equation of a DC generator.

Q11. Explain the types of DC generator and their applications.**Answer:**

A separately excited generator needs a separate field supply and is used for electroplating and electro-refining. A series generator is used as a booster and as a constant-current source for welding and arc lamps. A cumulative compound generator is used for lighting and long-distance transmission, while a differential compound generator is used for arc and welding work. A shunt generator is used for general lighting and power supply.

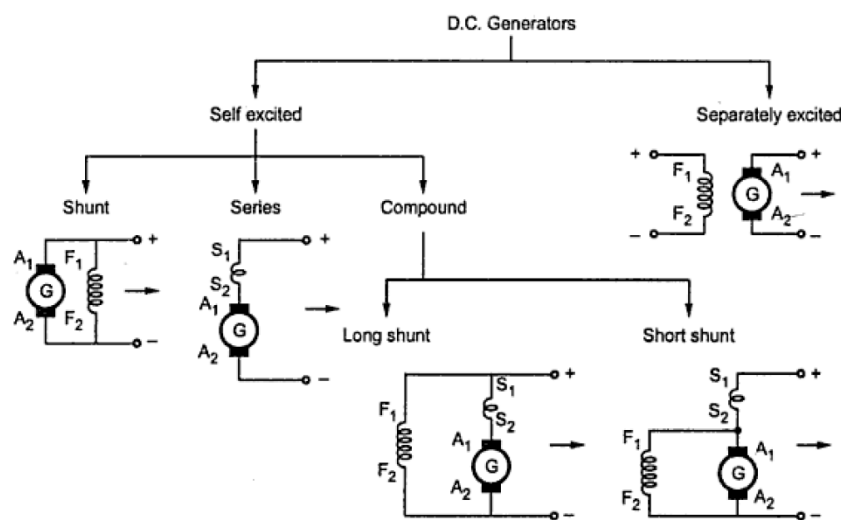


Figure 17: Types of DC generator connections.

Q12. Explain the force on a current-carrying conductor and Fleming's left-hand rule.**Answer:**

A conductor carrying current I , of active length l , placed at right angles to a field of flux density B experiences a force $F = BIl$. The direction is given by Fleming's left-hand rule: with the forefinger along the field and the middle finger along the current, the thumb gives the direction of the force. This is the basis of the DC motor.

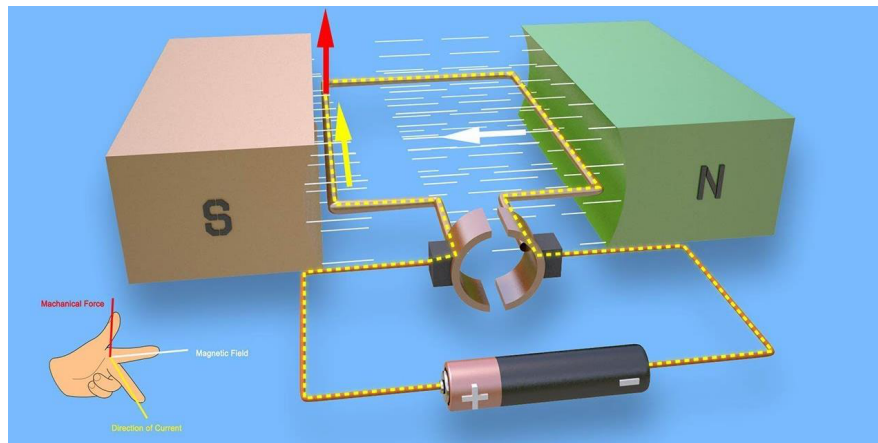


Figure 18: Force on a current-carrying conductor.

Q13. Explain the working principle of a DC motor, torque production and back emf.

Answer:

In a DC motor the current-carrying rotor conductors lie in the air-gap field and each experiences a force; since they lie at a common radius, a torque $T = 0.159 \phi Z I_a (P/A)$ is produced and the rotor rotates. The rotating armature also induces a back emf $E_b = \phi Z N P / (60 A)$ that opposes the supply, so that $V = E_b + I_a R_a$ and $E_b = V - I_a R_a$.

Type of Motor	Characteristics	Applications
Shunt	Speed is fairly constant and medium starting torque.	1. Blowers and fans 2. Centrifugal and reciprocating pumps 3. Lathe machines 4. Machine tools 5. Milling machines 6. Drilling machines
Series	High starting torque. No load condition is dangerous. Variable speed.	1. Cranes 2. Hoists, Elevators 3. Trolleys 4. Conveyors 5. Electric locomotives
Cumulative compound	High starting torque. No load condition is allowed.	1. Rolling mills 2. Punches 3. Shears 4. Heavy planers 5. Elevators
Differential compound	Speed increases as load increases.	Not suitable for any practical applications

Figure 19: DC motor and its parts.

Q14. Explain the applications of DC motors.

Answer:

Shunt motors are used for machine tools, lathes, blowers and fans; series motors are used for cranes, hoists, lifts and electric traction because they give a high starting torque; compound motors are used for rolling mills, presses, conveyors and shears.



Figure 20: Applications of DC motors.

SECTION C — Numerical Problems (Solved)

Every numerical below is solved step by step.

N1. Flux Density from Flux

Problem: A magnetic pole face has rectangular dimensions 200 mm × 100 mm. If the total flux emerging from the pole is 150 μWb, calculate the flux density.

Solution:

$$\text{Area } A = 0.2 \times 0.1 = 0.02 \text{ m}^2.$$

$$B = \phi/A = (150 \times 10^{-6})/0.02 = 7.5 \times 10^{-3} \text{ T}.$$

Answer: $B = 7.5 \text{ mT}$ ($7.5 \times 10^{-3} \text{ T}$).

N2. Radius of a Circular Pole Face

Problem: The maximum working flux density of a lifting electromagnet is 1.8 T and the total flux produced is 353 mWb. Determine the radius of the circular pole face.

Solution:

$$\text{Area } A = \phi/B = 0.353/1.8 = 0.1961 \text{ m}^2.$$

$$A = \pi r^2, \text{ so } r = \sqrt{A/\pi} = \sqrt{(0.1961/3.1416)} = 0.2499 \text{ m}.$$

Answer: $r \approx 0.25 \text{ m}$ (250 mm).

N3. Magnetic Field Strength and MMF of an Air-gap

Problem: Determine the magnetic field strength and the MMF required to produce a flux density of 0.25 T in an air-gap of length 12 mm.

Solution:

$$H = B/\mu_0 = 0.25/(4\pi \times 10^{-7}) = 198944 \text{ AT/m}.$$

$$\text{MMF} = H \times l = 198944 \times 0.012 = 2387 \text{ AT}.$$

Answer: $H = 198944 \text{ AT/m}$, $\text{MMF} = 2387 \text{ AT}$.

N4. Reluctance of a Metal Piece

Problem: Determine the reluctance of a piece of metal of length 150 mm and cross-sectional area 1800 mm^2 , if the relative permeability of the metal is 4000.

Solution:

$$S = l/(\mu_0 \mu_r A).$$

$$A = 1800 \times 10^{-6} = 1.8 \times 10^{-3} \text{ m}^2.$$

$$S = 0.15/(4\pi \times 10^{-7} \times 4000 \times 1.8 \times 10^{-3}) = 0.15/(9.048 \times 10^{-6}) = 16579 \text{ AT/Wb}.$$

Answer: $S = 16579 \text{ AT/Wb}$.

N5. Reluctance and Turns of a Mild Steel Ring

Problem: A mild steel ring has a radius of 50 mm and a cross-sectional area of 400 mm^2 . A current of 0.5 A flows in a coil wound uniformly around the ring and the flux produced is 0.1 mWb. If $\mu_r = 200$, find (a) the reluctance of the steel and (b) the number of turns.

Solution:

$$\text{Mean length } l = 2\pi r = 2\pi \times 0.05 = 0.3142 \text{ m}.$$

$$A = 400 \times 10^{-6} = 4 \times 10^{-4} \text{ m}^2.$$

$$(a) S = 0.3142/(4\pi \times 10^{-7} \times 200 \times 4 \times 10^{-4}) = 0.3142/(1.005 \times 10^{-7}) = 3.126 \times 10^6 \text{ AT/Wb}.$$

$$(b) \text{MMF} = \phi S = 0.1 \times 10^{-3} \times 3.126 \times 10^6 = 312.6 \text{ AT}; N = \text{MMF}/I = 312.6/0.5 = 625 \text{ turns}.$$

Answer: $S = 3.126 \times 10^6 \text{ AT/Wb}$, $N = 625 \text{ turns}$.

N6. Current from Magnetising Force

Problem: A magnetising force of 8000 A/m is applied to a circular magnetic circuit of mean diameter 30 cm by passing a current through a coil of 750 turns wound on the circuit. Find the current in the coil.

Solution:

$$\text{Mean length } l = \pi \times d = \pi \times 0.3 = 0.9425 \text{ m}.$$

$$\text{MMF} = H \times l = 8000 \times 0.9425 = 7540 \text{ AT}.$$

$$I = \text{MMF}/N = 7540/750 = 10.05 \text{ A}.$$

Answer: $I = 10.05 \text{ A}$.

N7. MMF and Field Strength of a Coil

Problem: A current of 5 A is passed through a 1000-turn coil wound on a circular magnetic circuit of radius 120 mm. Calculate (a) the magnetomotive force and (b) the magnetic field strength.

Solution:

$$(a) \text{MMF} = N I = 1000 \times 5 = 5000 \text{ AT}.$$

$$\text{Mean length } l = 2\pi r = 2\pi \times 0.12 = 0.754 \text{ m}.$$

$$(b) H = \text{MMF}/l = 5000/0.754 = 6632 \text{ AT/m}.$$

Answer: $\text{MMF} = 5000 \text{ AT}$, $H = 6632 \text{ AT/m}$.

N8. Composite Series Magnetic Circuit (Two Areas)

Problem: A closed magnetic circuit of cast steel contains a 6 cm long path of cross-sectional area 1 cm^2 and a 2 cm path of cross-sectional area 0.5 cm^2 . A coil of 200 turns is wound around the 6 cm length and a current of 0.4 A flows. Determine the flux density in the 2 cm path if μ_r of the cast steel is 750.

$$l = 2\pi r = 2 \times \pi \times 50 \times 10^{-3} \text{ m}; A = 400 \times 10^{-6} \text{ m}^2; I = 0.5 \text{ A}; \\ \Phi = 0.1 \times 10^{-3} \text{ Wb}; \mu_r = 200$$

$$(a) \text{ Reluctance } S = \frac{l}{\mu_0 \mu_r A} = \frac{2 \times \pi \times 50 \times 10^{-3}}{(4\pi \times 10^{-7})(200)(400 \times 10^{-6})} \\ = 3.125 \times 10^6 / \text{H}$$

Figure 21: Figure for N8.

Solution:

$$\text{MMF} = NI = 200 \times 0.4 = 80 \text{ AT.}$$

$$S_1 = 0.06 / (4\pi \times 10^{-7} \times 750 \times 1 \times 10^{-4}) = 6.366 \times 10^5 \text{ AT/Wb.}$$

$$S_2 = 0.02 / (4\pi \times 10^{-7} \times 750 \times 0.5 \times 10^{-4}) = 4.244 \times 10^5 \text{ AT/Wb.}$$

$$\text{Total } S = 1.061 \times 10^6 \text{ AT/Wb; flux } \phi = \text{MMF}/S = 80 / (1.061 \times 10^6) = 7.54 \times 10^{-5} \text{ Wb.}$$

$$B \text{ in the 2 cm path} = \phi / A_2 = 7.54 \times 10^{-5} / (0.5 \times 10^{-4}) = 1.51 \text{ T.}$$

Answer: $B = 1.51 \text{ T}$ in the 2 cm path.

N9. Magnetic Circuit with an Air-gap

Problem: A cast steel core with a uniform cross-sectional area of 2 cm^2 and a mean length of 25 cm has an air-gap 1 mm wide and a coil of 5000 turns. If μ_r of the steel is 48000, determine the current in the coil to produce a flux density of 0.80 T in the air-gap.

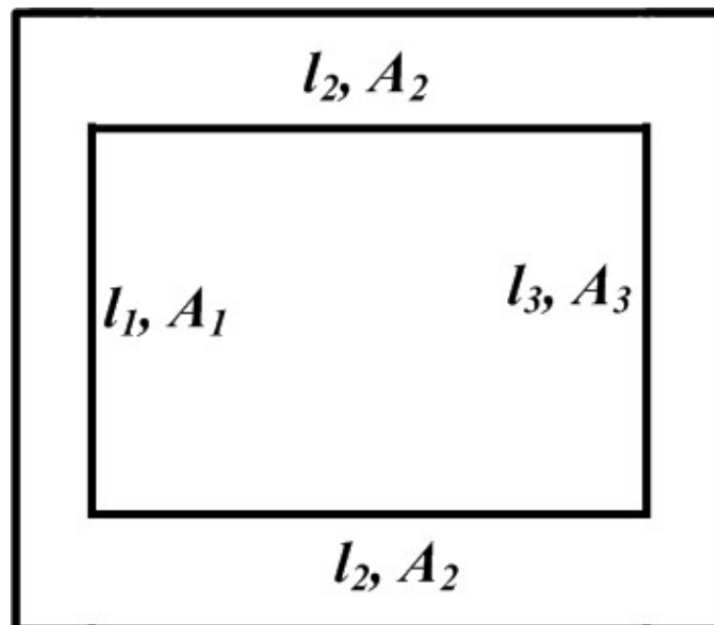


Figure 22: Figure for N9.

Solution:

$$\text{Flux } \phi = BA = 0.8 \times 2 \times 10^{-4} = 1.6 \times 10^{-4} \text{ Wb.}$$

$$S_{\text{steel}} = 0.25 / (4\pi \times 10^{-7} \times 48000 \times 2 \times 10^{-4}) = 2.072 \times 10^4 \text{ AT/Wb.}$$

$$S_{\text{gap}} = 0.001 / (4\pi \times 10^{-7} \times 2 \times 10^{-4}) = 3.979 \times 10^6 \text{ AT/Wb.}$$

$$\text{Total } S = 2.072 \times 10^4 + 3.979 \times 10^6 \approx 4.0 \times 10^6 \text{ AT/Wb.}$$

$$\text{MMF} = \phi S = 1.6 \times 10^{-4} \times 4.0 \times 10^6 = 640 \text{ AT; current } I = \text{MMF}/N = 640/5000 = 0.128 \text{ A.}$$

Answer: $I = 0.128 \text{ A.}$

N10. EMF Equation (Lap Winding)

Problem: A 4-pole DC generator has 500 armature conductors, a flux of 0.02 Wb per pole and runs at 1200 rpm. Find the generated emf for a lap winding ($A = P$).

$$\begin{aligned} &= \text{No. of slots} \times \text{No. of conductors/slot} \\ P &= \text{No. of generator poles} \\ A &= \text{No. of parallel paths in armature} \\ N &= \text{armature rotation in revolutions per minute (r.p.m.)} \\ E &= \text{e.m.f. induced in any parallel path in armature} \\ \text{Generated e.m.f. } E_g &= \text{e.m.f. generated in any one of the parallel paths i.e. } E. \\ \text{Average e.m.f. generated/conductor} &= \frac{d\Phi}{dt} \text{ volt } (\because n = 1) \\ \text{Now, flux cut/conductor in one revolution } d\Phi &= \Phi P \text{ Wb} \\ \text{No. of revolutions/second} &= N/60 \therefore \text{Time for one revolution, } dt = 60/N \text{ second} \\ \text{Hence, according to Faraday's Laws of Electromagnetic Induction,} \\ \text{E.M.F. generated/conductor} &= \frac{d\Phi}{dt} = \frac{\Phi P N}{60} \text{ volt} \end{aligned}$$

Figure 23: Figure for N10.

Solution:

$$E = (\phi Z N P) / (60 A).$$

For lap winding $A = P = 4$.

$$E = (0.02 \times 500 \times 1200 \times 4) / (60 \times 4) = 48000 / 240 = 200 \text{ V.}$$

Answer: $E = 200 \text{ V.}$

N11. EMF Equation (Wave Winding)

Problem: For the same machine as N10 (4 poles, 500 conductors, $\phi = 0.02 \text{ Wb}$, 1200 rpm), find the generated emf if the armature is wave-wound ($A = 2$).

Solution:

For wave winding $A = 2$.

$$E = (0.02 \times 500 \times 1200 \times 4) / (60 \times 2) = 48000 / 120 = 400 \text{ V.}$$

Answer: $E = 400 \text{ V}$ (double that of the lap winding).

N12. Back EMF of a DC Motor

Problem: A 220 V DC motor has an armature resistance of 0.5Ω and draws an armature current of 20 A. Find the back emf.

Solution:

$$V = E_b + I_a R_a, \text{ so } E_b = V - I_a R_a.$$

$$E_b = 220 - 20 \times 0.5 = 220 - 10 = 210 \text{ V.}$$

Answer: $E_b = 210 \text{ V.}$

N13. Torque of a DC Motor

Problem: A 4-pole DC motor has 400 armature conductors, a flux of 0.03 Wb per pole and an armature current of 25 A. The armature is lap-wound ($A = 4$). Find the torque developed.

Solution:

$$T = 0.159 \phi Z I_a \text{ (P/A)}.$$

$$T = 0.159 \times 0.03 \times 400 \times 25 \times (4/4) = 47.7 \text{ N}\cdot\text{m}.$$

Answer: $T = 47.7 \text{ N}\cdot\text{m}.$